



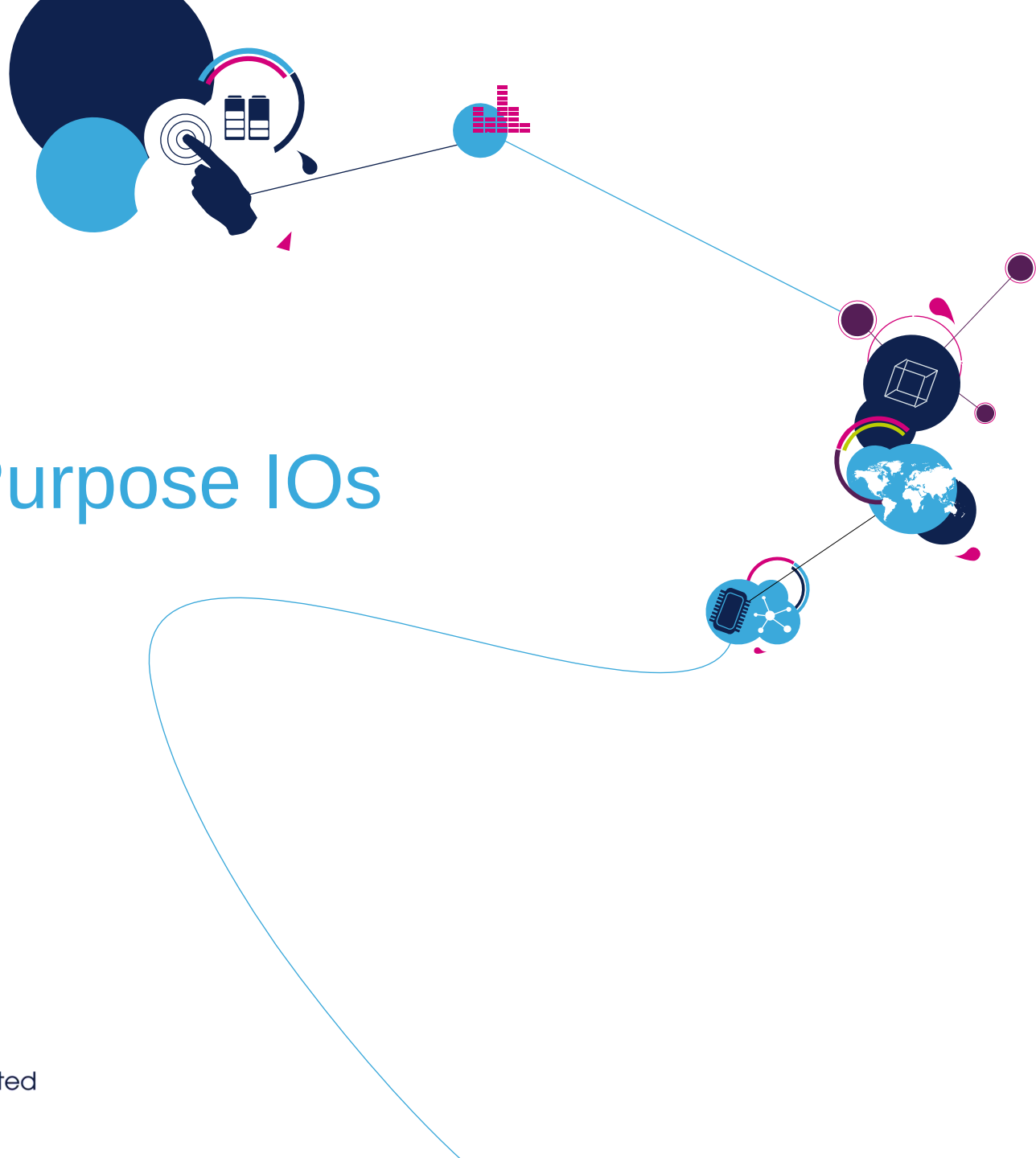
STM32F3 Technical Training

For reference only

Refer to the latest documents for details



General Purpose IOs

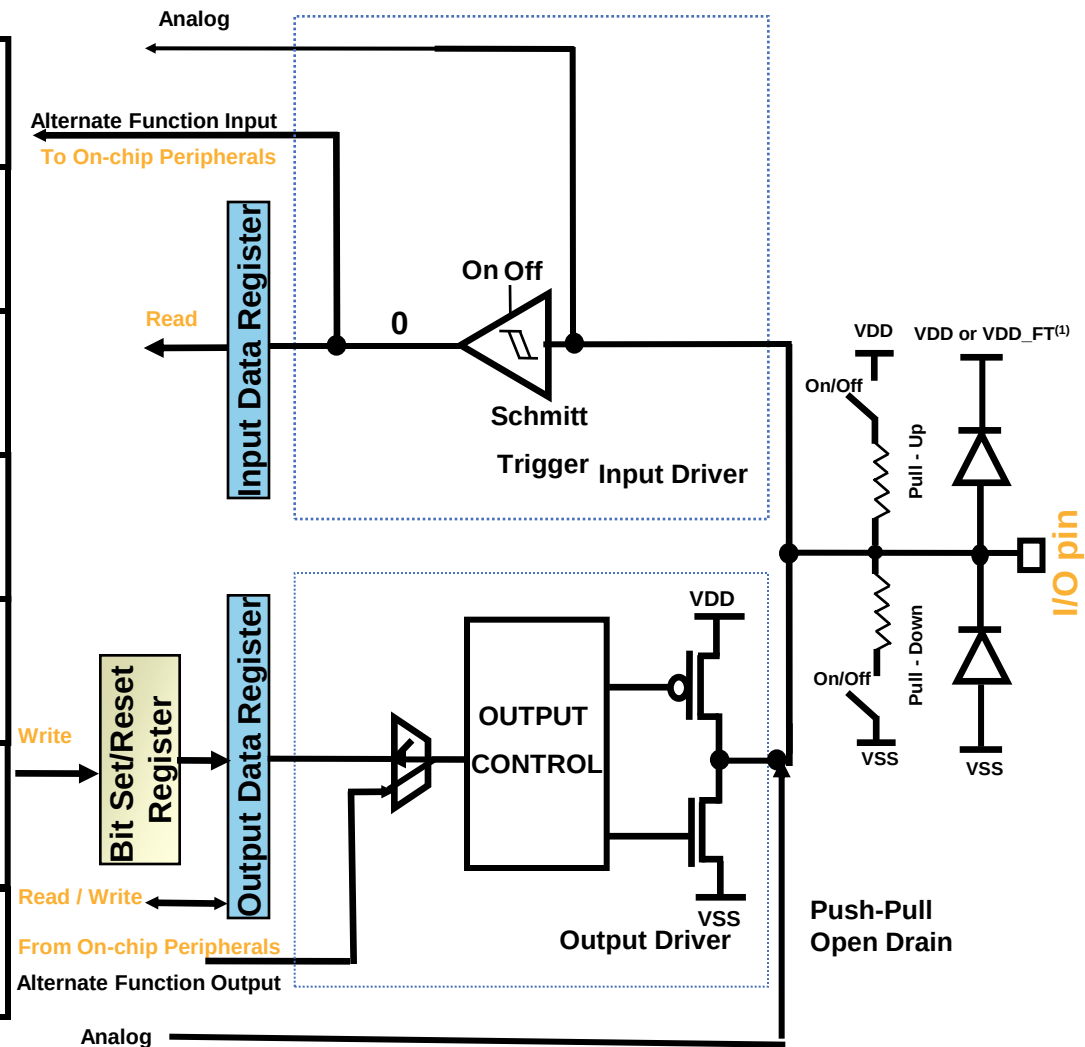


- Up to 84 (in STM32F37x) and 87 (in STM32F30x) multifunction bi-directional I/O ports available on biggest package 100 pin.
- Several I/Os are 5V tolerant (ADC, opamp, comparators pins are not).
- All Standard I/Os are shared in 6 ports: GPIOA, GPIOB, GPIOC, GPIOD, GPIOE, GPIOF.
- Atomic Bit Set and Bit Reset using BSRR and BRR registers
- GPIO connected to AHB bus, max toggling frequency 18 MHz
- Configurable Output slew rate speed up to 50MHz
- Locking mechanism (GPIOx_LCKR) provided to freeze the I/O configuration
 - When the LOCK sequence has been applied on a port bit, it is no longer possible to modify the configuration of the port bit until the next reset (no write access to the CRL and CRH registers corresponding bit).
- Up to 84 (in STM32F37x) and 87 (in STM32F30x) GPIOs can be set-up as external interrupt (up to 16 lines at time) able to wake-up the MCU from low power modes.

GPIO Configuration Modes

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MODER(i) [1:0]	OTYPER(i) [1:0]	PUPDR(i) [1:0]	I/O configuration
01	0	0 0 0 1 1 0	Output Push Pull Output Push Pull with Pull-up Output Push Pull with Pull-down
	1	0 0 0 1 1 0	Output Open Drain Output Open Drain with Pull-up Output Open Drain with Pull-down
10	0	0 0 0 1 1 0	Alternate Function Push Pull Alternate Function PP Pull-up Alternate Function PP Pull-down
	1	0 0 0 1 1 0	Alternate Function Open Drain Alternate Function OD Pull-up Alternate Function OD Pull-down
00	x	0 0 0 1 1 0	Input floating Input with Pull-up Input with Pull-down
11	x	x	Analog mode



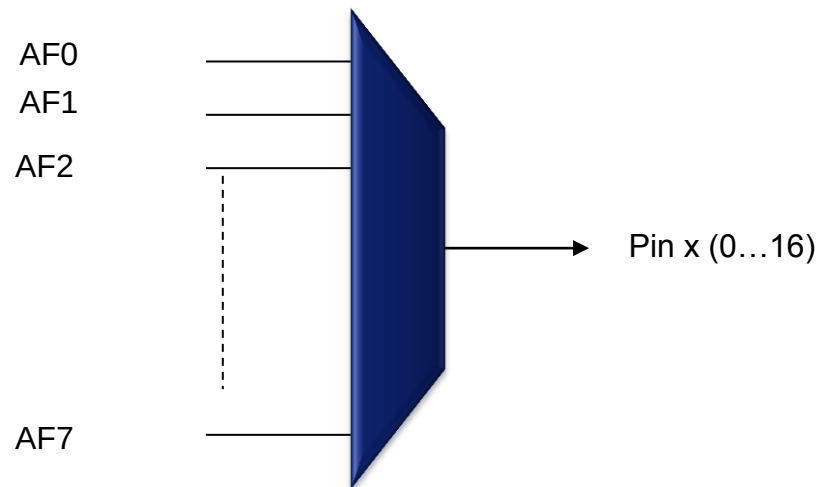
* In output mode, the I/O speed is configurable through OSPEEDR register: 2MHz, 10MHz or 50MHz

(1) VDD_FT is a potential specific to five-volt tolerant I/Os and different from VDD.

Alternate Functions features

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- Most of the peripherals shares the same pin (like USARTx_TX, TIMx_CH2, I2Cx_SCL, SPIx_MISO, EVENTOUT...)
- Alternate functions multiplexers prevent to have several peripheral's function pin to be connected to a specific I/O at a time.

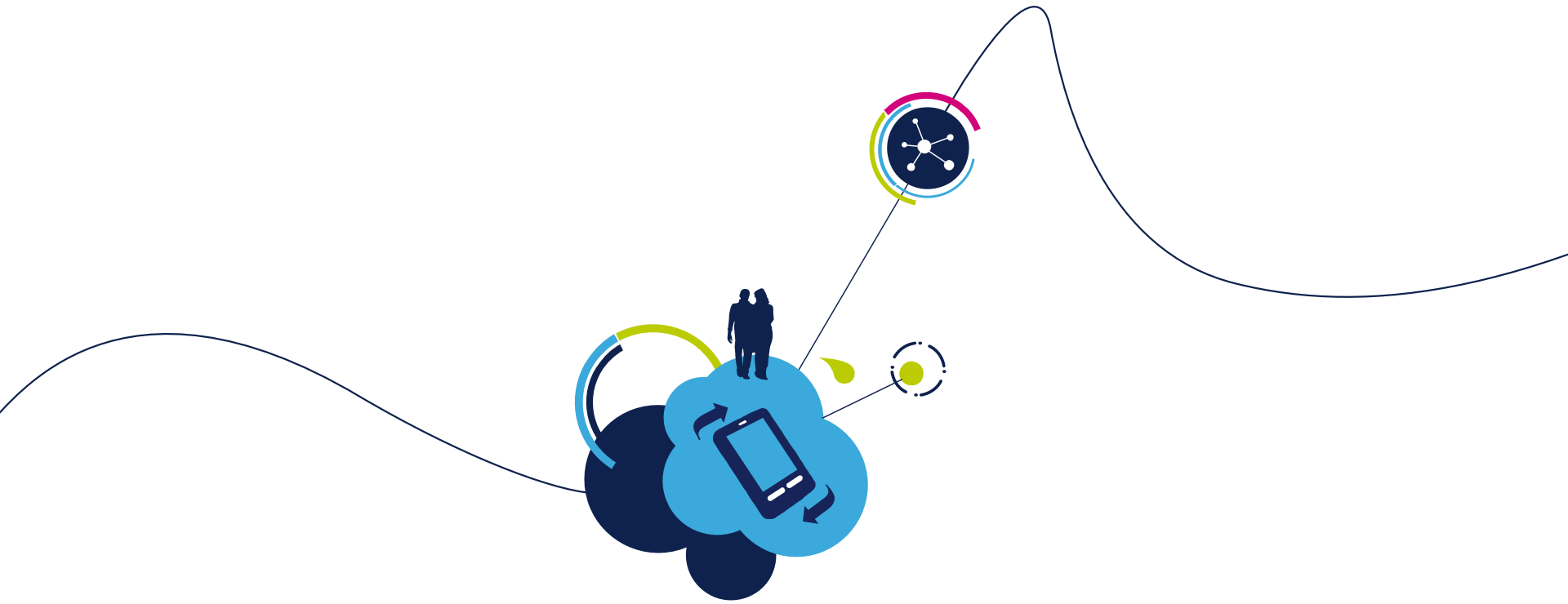


- During and just after reset, the alternate functions are not active and the I/O ports are configured in input floating mode. But, the debug pins (JTAG/SWD) are in AF pull-up/pull-down after reset:
 - PA13: JTMS/SWDIO
 - PA14: JTCK/SWCLK
 - PA15: JTDI
 - PB3: JTDO
 - PB4: NJTRST
- Using the HSE or LSE oscillator pins as GPIOs
 - When the HSE or LSE oscillator is switched OFF (default state after reset), the related oscillator pins can be used as general purpose I/Os.
 - When the oscillator is configured in a user external clock mode, only the OSC_IN or OSC32_IN pin is reserved for clock input and the OSC_OUT or OSC32_OUT pin can still be used as general purpose I/Os.
- Using the GPIO pins in the backup supply domain
 - The PC13/PC14/PC15 GPIO functionality is lost when the device enters Standby mode. In this case, if their GPIO configuration is not bypassed by the RTC configuration, these pins are set in an analog input mode.

- How many I/Os and ports there are in the STM32F3xx microcontroller ?

- List all the I/O configuration modes

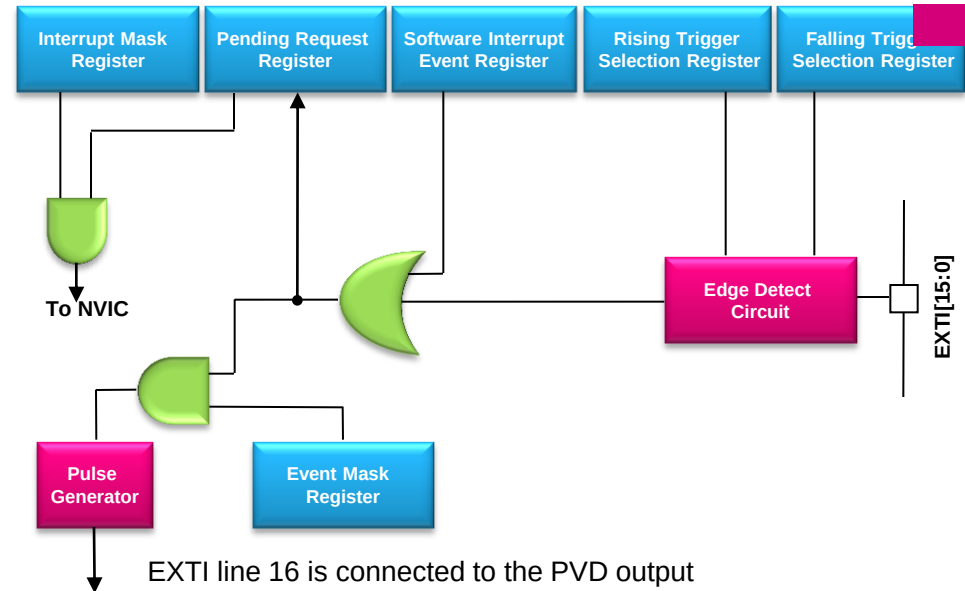
- How many External interrupts and Wake-up pins, exist in the STM32F3xx microcontroller?



Extended interrupts and events controller (EXTI)

EXTI Features

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EXTI line 16 is connected to the PVD output
 EXTI line 17 is connected to the RTC Alarm event
 EXTI line 18 is connected to USB Device FS wakeup event
 EXTI line 19 is connected to RTC tamper and Timestamps
 EXTI line 20 is connected to RTC wakeup
 EXTI line 21 is connected to Comparator 1 output
 EXTI line 22 is connected to Comparator 2 output
 EXTI line 23 is connected to I2C1 wakeup
 EXTI line 24 is connected to I2C2 wakeup
 EXTI line 25 is connected to USART1 wakeup
 EXTI line 26 is connected to USART2 wakeup.
 EXTI line 27 is connected to CEC wakeup. (STM32F37x only)
 EXTI line 28 is connected to USART3 wakeup
 EXTI line 29 is connected to Comparator 3 output (F30x only)
 EXTI line 30 is connected to Comparator 4 output (F30x only)
 EXTI line 31 is connected to Comparator 5 output. (F30x only)
 EXTI line 32 is connected to I Comparator 6 output (F30x only)
 EXTI line 33 is connected to Comparator 7 output (F30x only)
 EXTI line 34 is connected to UART4 wakeup. (F30x only)
 EXTI line 35 is connected to UART5 wakeup. (F30x only)

- Manages the external and **internal asynchronous** events/interrupts and generates the event request to the CPU/Interrupt Controller and a wake-up request to the Power Manager
- Some communication peripherals (UART, I2C, CEC (*), comparators) are able to generate events when the system is in run/sleep mode and also when the **system is in stop mode** allowing to wake up the system from stop mode.
- These peripherals are able to generate both a synchronized (to the system APB clock) and an asynchronous version of the event.
- All others features are same as STM32F1xx series
- Up to 36 (F30x) 29(F37x) Interrupt/Events requests : Up to 88 (in STM32F30x) and 84 (in STM32F37x) GPIOs can be used as EXTI line(0..15)
- (*) The CEC is available on STM32F37x only.

- How many lines does the Extended interrupt controller support?

- Which lines are mapped to a special asynchronous events ?

- Which lines can be used as system wake-up ?



Thank you !

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